

Graph-Based Intrusion Detection: Detecting Network Attacks with Graph Neural Networks and LLM-Generated Alert Explanations

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Abstract—Traditional machine-learning-based network intrusion detection systems (NIDS) typically classify each network flow independently, ignoring communication structure among hosts. However, attacks such as scanning, distributed denial-of-service, and lateral movement are often defined by patterns involving multiple hosts rather than by a single flow. This project proposes a graph-based intrusion detection framework in which hosts are modeled as nodes and network flows as attributed edges. We will construct time-windowed communication graphs from the standardized NF-UNSW-NB15-v2 and NF-CSE-CIC-IDS2018-v2 datasets and implement an edge-aware GNN based on E-GraphSAGE. The model will be compared with Random Forest, XGBoost, and multilayer perceptron baselines. We will also investigate attention-based aggregation and cross-dataset generalization using binary benign-versus-malicious classification. Finally, an LLM explanation layer will convert detected malicious flows and their graph context into short human-readable alerts. Evaluation will emphasize macro and per-class F1 because of class imbalance.

Index Terms—intrusion detection, graph neural networks, network security, deep learning, edge classification, domain generalization, large language models

I. INTRODUCTION

Machine-learning-based network intrusion detection systems commonly represent each network flow as an independent feature vector containing information such as duration, packet counts, byte counts, protocol, and port numbers. Although effective, this formulation does not explicitly model relationships among network hosts.

This limitation is important because many attacks are characterized by communication patterns rather than by a single connection. For example, hundreds of connections from one source to many destinations may indicate scanning or brute-force activity, while many sources simultaneously targeting one destination may indicate a denial-of-service attack. Such behaviors naturally form graph structures.

We therefore formulate intrusion detection as an edge-classification problem. For each time window t , we construct a directed multigraph $G_t = (V_t, E_t)$, where V_t contains observed hosts and each $e \in E_t$ represents a network flow with feature vector $\mathbf{x}_e$. The goal is to predict whether each

edge is benign or malicious and, for in-domain experiments, its attack category.

The goal of this project is to develop and evaluate a graph-based intrusion detection framework that combines edge-aware graph learning, cross-dataset generalization, and human-readable alert explanations. Our project investigates four questions: whether graph-based modeling can outperform flow-level classifiers; whether attention-based aggregation improves over mean aggregation; how well the learned model generalizes across datasets; and whether an LLM can convert predictions and graph context into useful human-readable alerts.

II. RELATED WORK

Traditional intrusion detection methods commonly classify individual flows using tabular features, with ensemble methods such as Random Forest and XGBoost remaining strong baselines. However, these approaches largely treat flows independently and therefore do not explicitly model relationships among communicating hosts.

Graph neural networks provide a natural alternative because network traffic can be represented as a communication graph. GraphSAGE [3] learns inductive neighborhood aggregation functions, while graph attention networks [4] allow learned weighting of neighboring information. E-GraphSAGE [1] adapts GraphSAGE for network intrusion detection by aggregating incident edge features and classifying network flows using endpoint embeddings. Anomal-E [2] further demonstrates the usefulness of graph-based edge representations for intrusion detection.

A major limitation of existing graph-based NIDS research is that evaluation is often restricted to a single dataset. Cross-dataset comparison is difficult because intrusion datasets typically use different feature schemas. The standardized NetFlow datasets introduced by Sarhan et al. [5] provide a common representation across multiple datasets and therefore enable direct cross-dataset evaluation. Our project combines this standardized representation with edge-centric graph learning, leakage-aware evaluation, and LLM-generated alert explanations.

III. DATASET AND PREPROCESSING

We use two datasets from the standardized NetFlow NIDS collection [5]: NF-UNSW-NB15-v2, derived from UNSW-NB15 [6], and NF-CSE-CIC-IDS2018-v2, derived from CSE-CIC-IDS2018 [7]. Both use the same extended NetFlow feature schema, allowing direct cross-dataset evaluation without feature re-engineering.

Flows will be partitioned into 60-second time windows. Within each window, each unique IP address becomes a node and each network flow becomes a directed edge carrying its flow features. IP addresses will be used only for graph construction and excluded from predictive features to reduce identifier leakage.

Exact duplicate records will be removed, selected numerical features will be transformed and standardized using statistics fitted only on the training data, and categorical protocol-related features will be encoded consistently across splits. Where reliable timestamps are available, chronological or episode-aware splitting will be used to reduce leakage between training and testing data.

Because the datasets are strongly imbalanced, class-weighted loss will be used where appropriate, and evaluation will emphasize macro and per-class F1 rather than accuracy alone.

IV. PROPOSED APPROACH

A. Baselines

To evaluate whether graph structure provides additional predictive value beyond flow-level features, we plan to compare the proposed GNN against three non-graph baselines trained using the same input features, data splits, and class-weighting strategy:

- 1) Random Forest.
- 2) XGBoost.
- 3) A multilayer perceptron (MLP) with a parameter count chosen to be reasonably comparable to the GNN.

B. Graph Neural Network Model

Our primary graph model is based on the E-GraphSAGE formulation, which is designed for settings in which informative features are associated primarily with edges rather than nodes. At each message-passing layer, a node updates its representation by aggregating information from the incident network-flow edges:

$$h_v^{(k)} = \sigma \left(W^{(k)} \cdot \text{CONCAT} \left(h_v^{(k-1)}, \text{AGG}_{u \in N(v)} \left(h_{uv}^{(k-1)} \right) \right) \right) \quad (1)$$

where h_{uv} denotes the representation associated with the flow edge between nodes u and v , and AGG denotes a permutation-invariant aggregation function such as mean aggregation.

After K message-passing layers, the representation used for edge classification is constructed by concatenating the learned embeddings of the source and destination nodes with the original flow features:

$$z_{uv} = \text{CONCAT} \left(h_u^{(K)}, h_v^{(K)}, x_{uv} \right) \quad (2)$$

The resulting edge representation is then passed to a classification head that outputs a probability distribution over the benign class and the available attack categories.

C. Proposed Extension

Beyond the baseline E-GraphSAGE model, we plan to investigate an attention-based aggregation mechanism. Instead of assigning equal importance to all neighboring flows during message passing, the attention mechanism learns weights that allow the model to emphasize more informative or suspicious connections.

This extension is motivated by the observation that network hosts may participate in many benign connections while only a small subset of their neighboring flows are associated with malicious behavior. Mean aggregation may dilute these important signals, whereas attention-based aggregation may allow the model to focus more strongly on the most relevant communication patterns.

We will compare attention-based aggregation against the original mean aggregation used in the baseline model to evaluate whether learned neighbor weighting improves intrusion detection performance, particularly for minority attack classes.

D. LLM Explanation Layer

As an extension to the core intrusion detection pipeline, we plan to add a large language model (LLM) explanation layer that converts model predictions into short, human-readable security alerts. The LLM will not participate in the detection decision itself; instead, it will receive structured information derived from the GNN output and the corresponding local graph context.

For each flow predicted as malicious, the explanation input will include the predicted attack class, prediction confidence, selected flow features, and local graph statistics such as source-host fan-out, the number of distinct destination ports contacted, and the number of similar flows observed within the same time window. This information will be passed to an LLM using a fixed prompt template through the Virginia Tech ARC LLM Gateway.

The generated explanation will summarize why the flow was flagged in plain language while remaining grounded in the supplied model and traffic evidence. To reduce unsupported explanations, the LLM will be instructed to use only the information provided in the structured input and not introduce additional attack evidence. The explanation component will be evaluated qualitatively for factual consistency and clarity.

E. Evaluation Protocol

- **Primary metrics:** macro F1 and per-class F1.
- **Secondary metrics:** precision, recall, confusion matrices, and precision–recall AUC where appropriate.
- **In-domain evaluation:** held-out train/validation/test splits within each dataset, using chronological splitting where reliable timestamps are available.

- **Cross-dataset evaluation:** train on NF-UNSW-NB15-v2 and test on NF-CSE-CIC-IDS2018-v2, and vice versa, using a binary benign-versus-malicious formulation to avoid inconsistencies between the attack taxonomies of the two datasets.
- **Ablation studies:** we prioritize two comparisons: graph-based modeling versus flow-level modeling, and attention-based aggregation versus mean aggregation. Additional studies on time-window length and class weighting will be conducted if time permits.
- **Qualitative evaluation:** manual review of a sample of LLM explanations for factual consistency with the underlying flow.

V. SYSTEM DESIGN

The system consists of four stages: preprocessing and graph construction, model training, inference and explanation, and visualization. Models will be implemented using PyTorch and PyTorch Geometric and trained using Virginia Tech computing resources.

The overall pipeline is:

NetFlow Data → Graph Construction → GNN/Baselines
→ Attack Prediction → LLM Explanation

If time permits, a lightweight dashboard will visualize selected communication graphs, malicious-flow predictions, confidence scores, evaluation results, and LLM-generated explanations.

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TABLE I
TASK ASSIGNMENT

Member	Responsibilities
Shiyi Liu	Dataset preparation, preprocessing, temporal splitting, and graph construction.
Vighnesh Nair	Random Forest, XGBoost, MLP baselines, evaluation metrics, and ablation support.
Krishna Inukonda	E-GraphSAGE, attention-based extension, hyperparameter tuning, and cross-dataset experiments.
Gokulakrishnan Ravi	LLM explanation module, dashboard, project website, and demonstration support.
All members	Proposal/report writing and checkpoint/final presentations.

TABLE II
PROJECT SCHEDULE

Period	Milestone
Sep 15–Oct 6	Dataset preprocessing, graph construction, and baseline models.
Oct 7–20	Initial E-GraphSAGE implementation and evaluation.
Oct 21–Nov 10	Attention extension, cross-dataset experiments, and LLM prototype.
Nov 11–Dec 11	Ablation studies, dashboard, final experiments, presentation, and final report.